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Table 29
Cyclic Oxidation Resistance

Cycle	Alloy	Weight Change, mg/cm ²					
		134 cycles	275 cycles	467 cycles	200 cycles	304 cycles	400 cycles
1600-1700 F (871-927 C)	RA 330	+ 3.4	+ 4.9	+ 6.4	—	—	—
25 minutes heat	Type 310	+ 4.0	+ 6.7	22.7	—	—	—
5 minutes cool	Type 309	+ 3.0	41.6	100.4	—	—	—
duplicate tests	NITRONIC 60	+ 1.5	69.2	167.6	—	—	—
		Weight Loss, mg/cm ²					
1900 F (1038 C)	Type 446	—	—	—	1.47	1.72	1.97
30 minutes heat,	Type 310	—	—	—	2.70	15.95	17.22
30 minutes cool	Type 309	—	—	—	22.53	26.34	33.69
	NITRONIC 60	—	—	—	42.99	60.40	74.80
	Type 316	—	—	—	93.04	135.34	178.27

Mechanical Properties

Table 30
Typical Room Temperature Tensile Properties* (See Table 36 for acceptable specification values.)

Condition	Size	Hardness	UTS ksi (MPa)	0.2% YS ksi (MPa)	Elongation % in 4XD	Reduction of Area, %
Annealed	1" (25.4 mm) Ø	95 HRB	103 (710)	60 (414)	64	74
Annealed	1-3/4" (44.4 mm) Ø	100 HRB	101 (696)	56 (386)	62	73
Annealed	2-1/4" (57.2 mm) Ø	100 HRB	101 (696)	60 (414)	60	76
Annealed	3" (76.2 mm) Ø	97 HRB	113 (779)	65 (448)	55	67
Annealed	4-1/8" (104.8 mm) Ø	95 HRB	106 (731)	56 (386)	57	67
AMS 5848			95 min.	50 min.		1/2" over Ø
AMS 5898			100 min.	55 min.		under 1/2" Ø

*Data based on duplicate tests
(1) CG bar

Table 31
Typical Bearing Properties
ASTM E 238

Condition	Bearing Strength ksi (MPa)	Bearing Yield Strength ksi (MPa)	0.2% YS ksi (MPa)	UTS ksi (MPa)	% El in 2"	Hardness (R)
Annealed	190.5 (1313)	79.5 (548)	104.9 (723)	52.2 (360)	49.2	B90
10% Cold Rolled	212 (1462)	132.8 (916)	123.1 (849)	90.6 (625)	40.0	C26



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